ShopEase Authentication Service

FastAPI Authentication Microservice with JWT, MySQL and Docker

Submitted By: Dineshkumar S

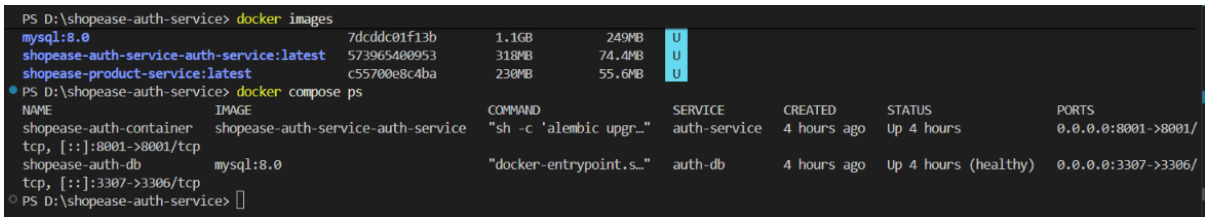
Date: 05/11/2026

Requirement Completion

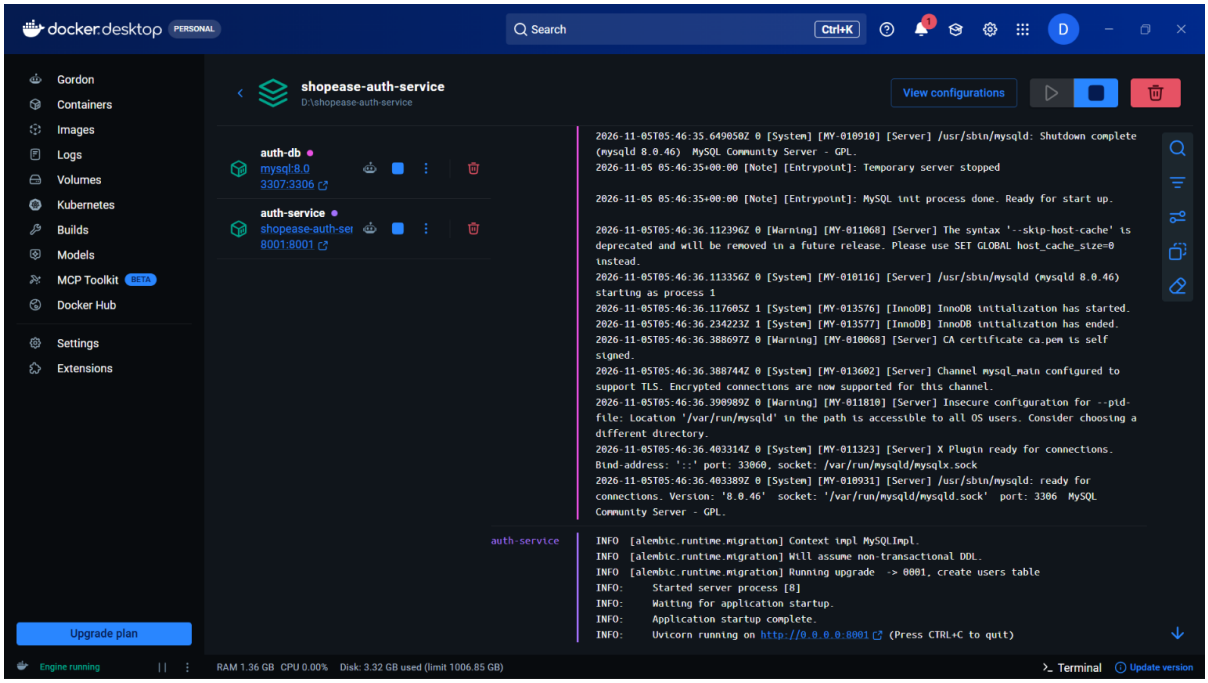
Requirement	Status	Evidence
Authentication Service Source Code	Done	FastAPI project with separate folders for routers, services, models, schemas, core and database. main.py only starts the app and registers the router.
MySQL Database	Done	Separate MySQL 8 database shopease_auth_db in its own container, with a users table defined by a SQLAlchemy model.
Alembic Migration	Done	Migration 0001 creates the users table and runs automatically when the container starts.
User Registration API	Done	POST /api/auth/register validates email, password strength and matching passwords, rejects duplicate email (409), hashes the password and assigns the role user.
User Login API	Done	POST /api/auth/login checks the bcrypt hash, rejects wrong credentials (401) and inactive users (403), and returns a JWT with user details.
JWT Authentication	Done	Token holds user ID, role and expiry. Secret, algorithm and expiry come from .env. Expired or invalid tokens return 401.
Get Current User API	Done	GET /api/auth/me returns the profile for a valid Bearer token.
Verify Token and Change Password APIs	Done	POST /api/auth/verify-token validates a token. POST /api/auth/change-password needs a valid token and the correct current password (400 if wrong).
Dockerfile	Done	Based on python:3.11-slim, installs requirements.txt, exposes port 8001, runs the migration and starts Uvicorn.
Docker Compose	Done	Runs auth-service and auth-db (mysql:8.0) with a named volume, a MySQL health check and MYSQL_HOST set to the service name auth-db.
Environment Variables and Secrets	Done	Database credentials and JWT secret are in .env, which is excluded by .gitignore and .dockerignore.
API Testing	Done	All 14 mandatory scenarios tested in Swagger UI with correct status codes, plus verify-token. Also run in Bruno using the Postman collection.
README Documentation	Done	Overview, folder structure, setup, Alembic and Docker commands, endpoints, JWT steps and test results.
GitHub Repository	Done	Source code pushed to https://github.com/Dineshkumarsenthil/shopease-auth-service

Evidence

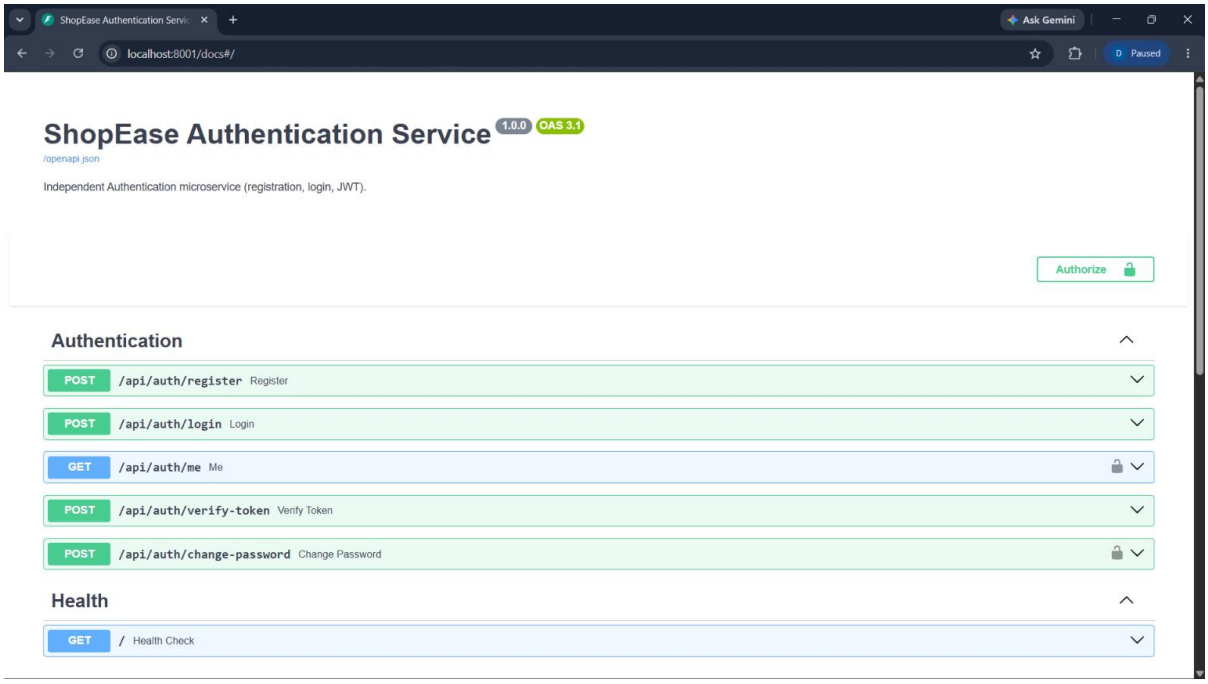
Docker: Containers Running (docker images, docker compose ps)



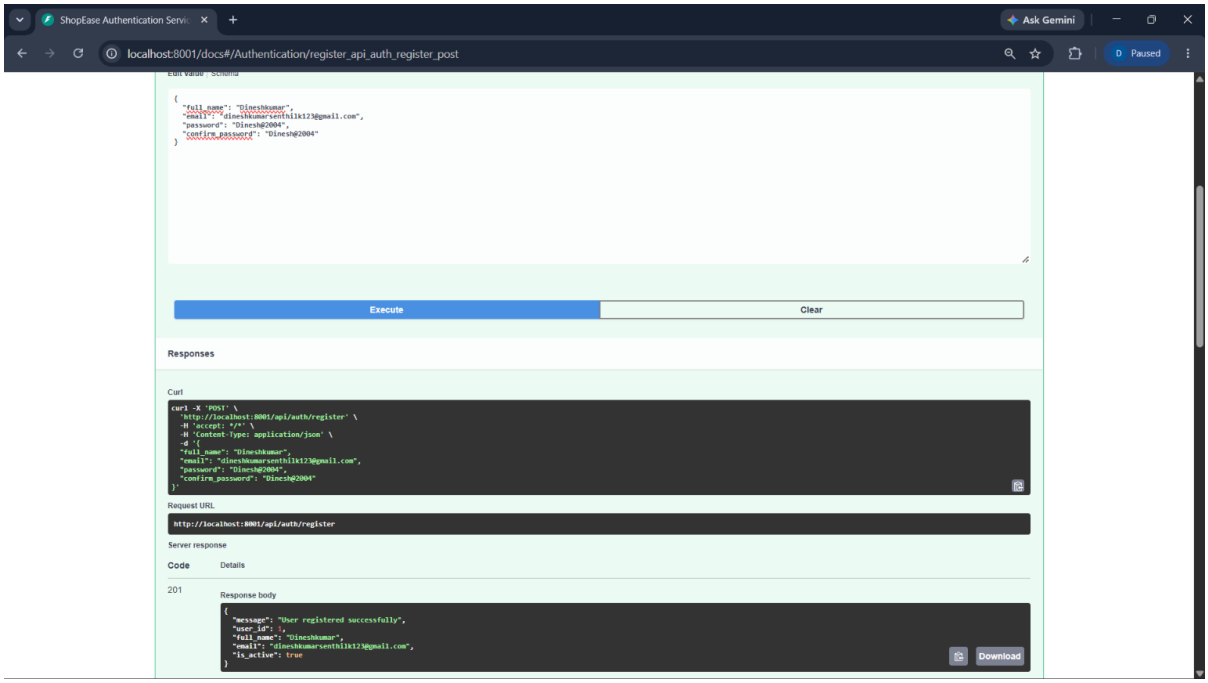
Docker: Logs, MySQL Ready, Alembic Migration and Uvicorn Running on Port 8001



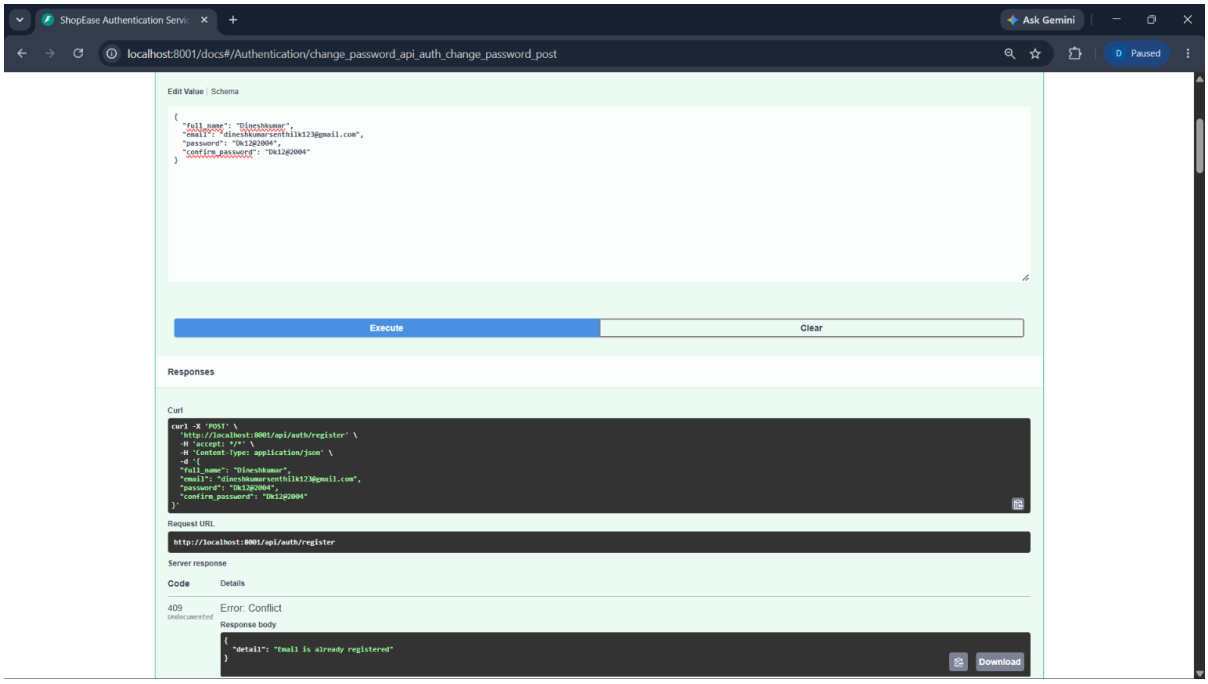
API Documentation: Swagger UI Endpoint List



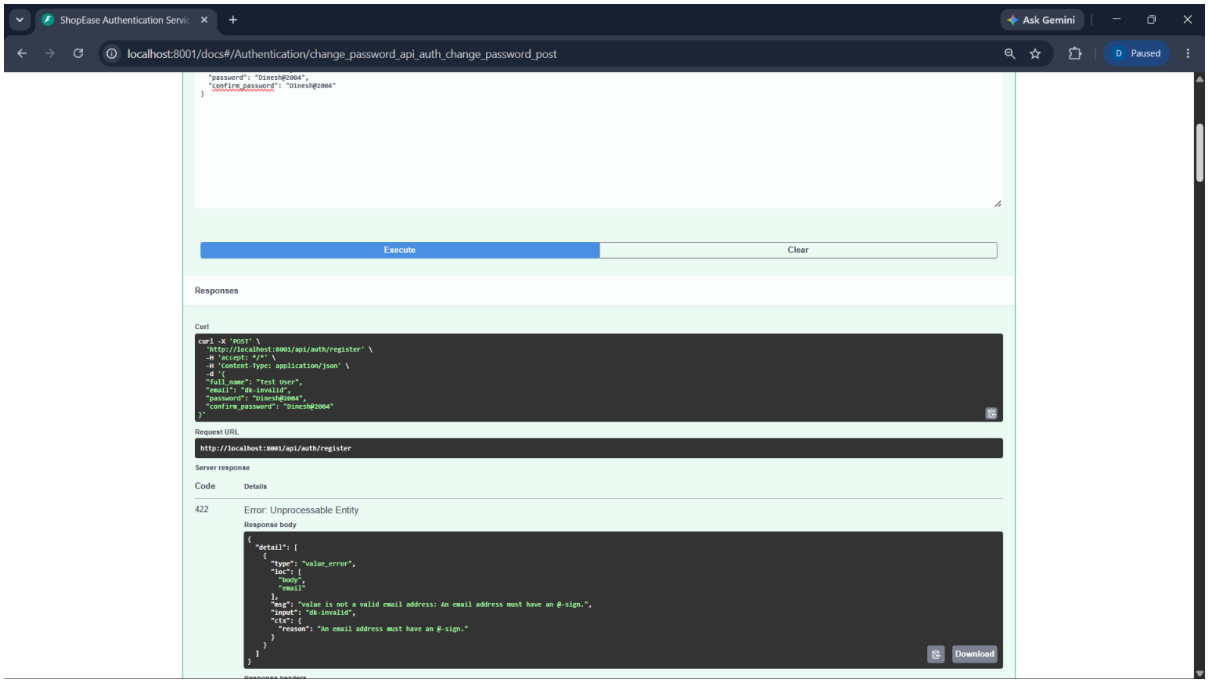
Scenario 1: Register a New User, POST /api/auth/register, 201 Created



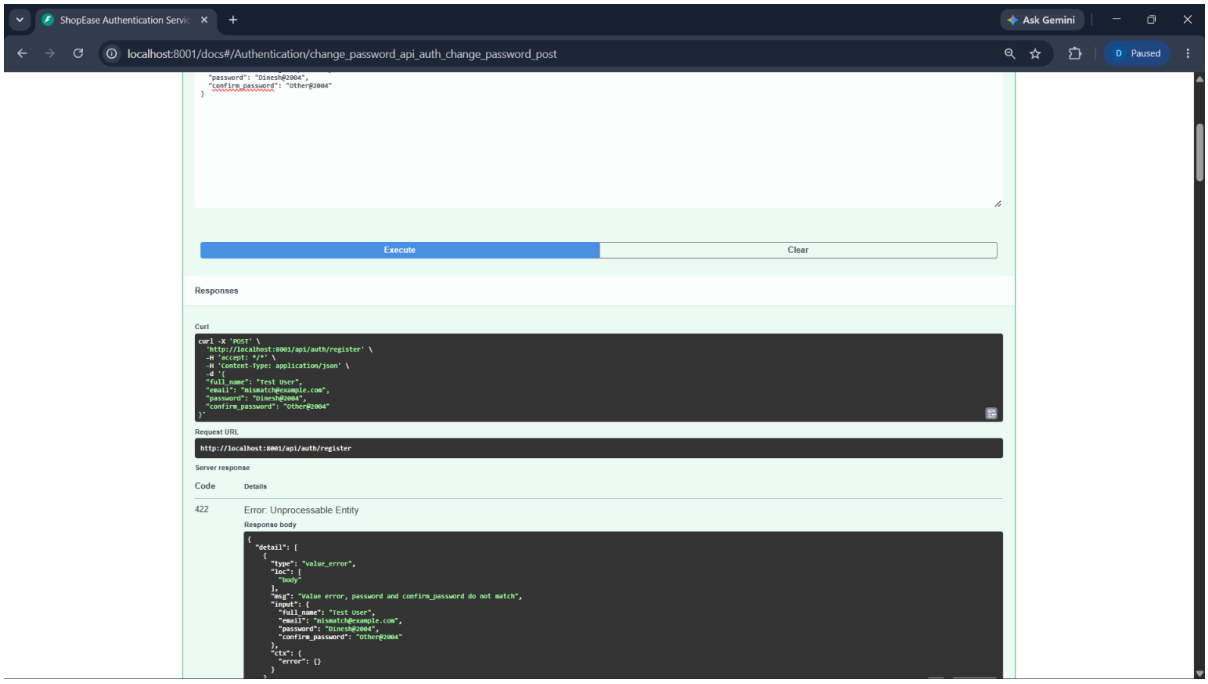
Scenario 2: Duplicate Email, 409 Conflict



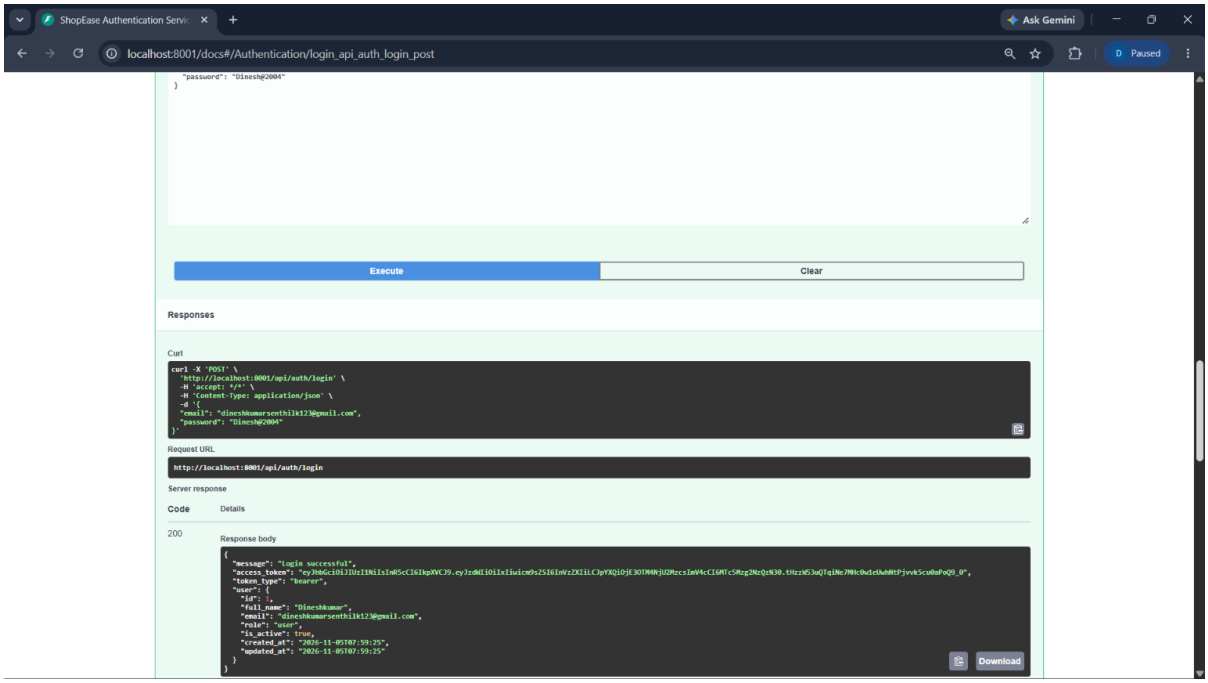
Scenario 3: Invalid Email, 422 Unprocessable Entity



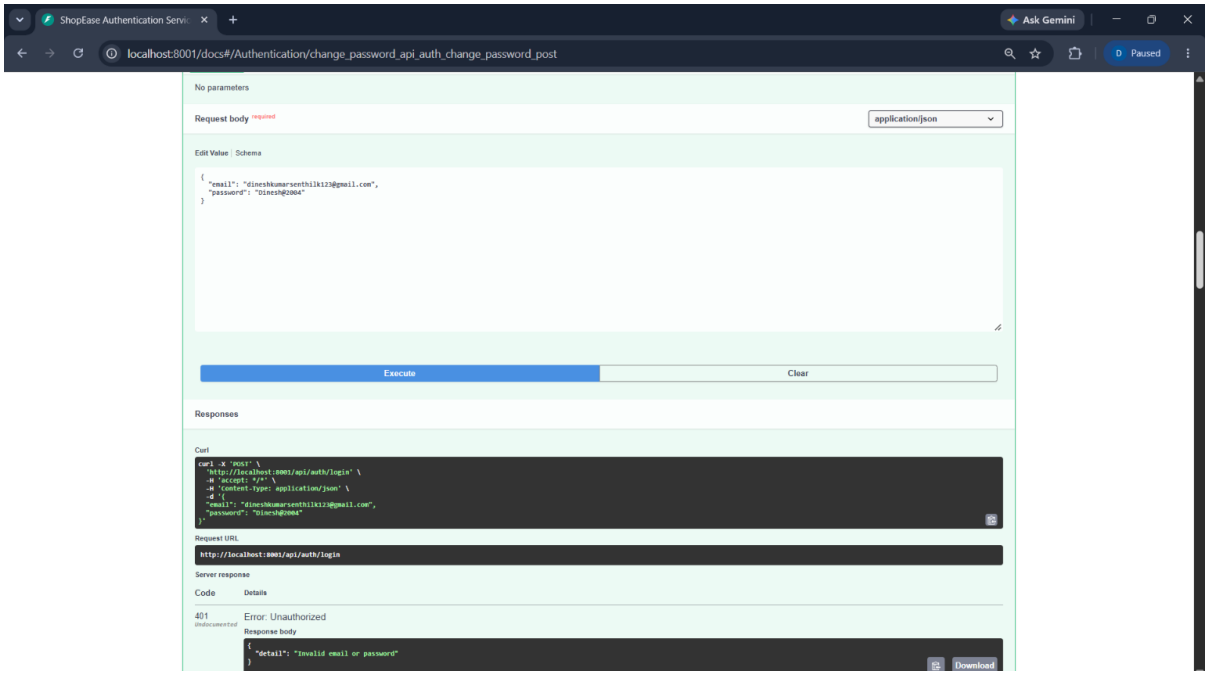
Scenario 4: Mismatched Passwords, 422 Unprocessable Entity



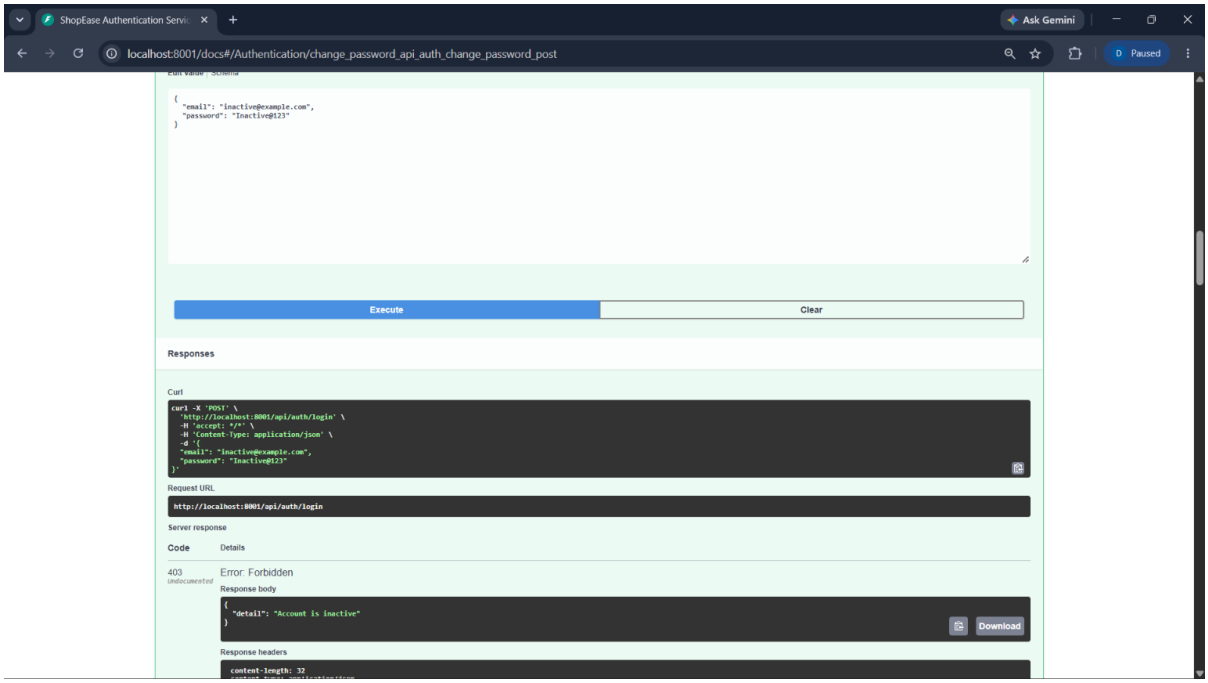
Scenario 5: Login with Valid Credentials, 200 OK



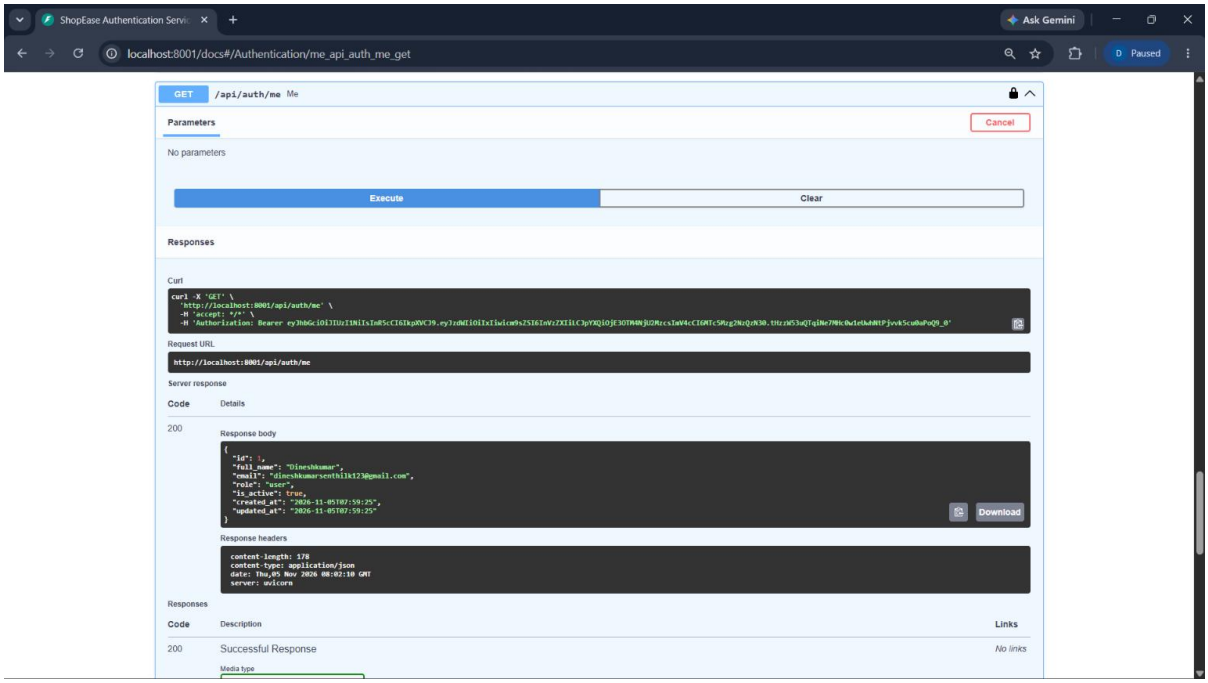
Scenario 6: Login with Incorrect Password, 401 Unauthorized



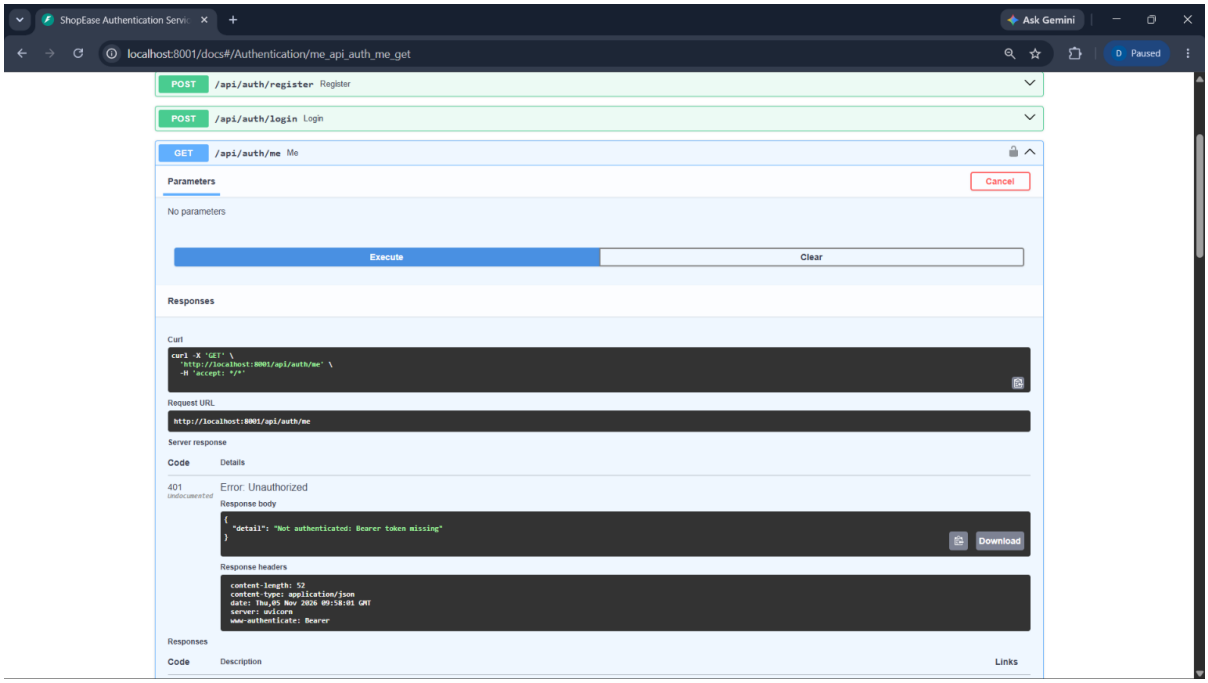
Scenario 7: Login with Inactive Account, 403 Forbidden



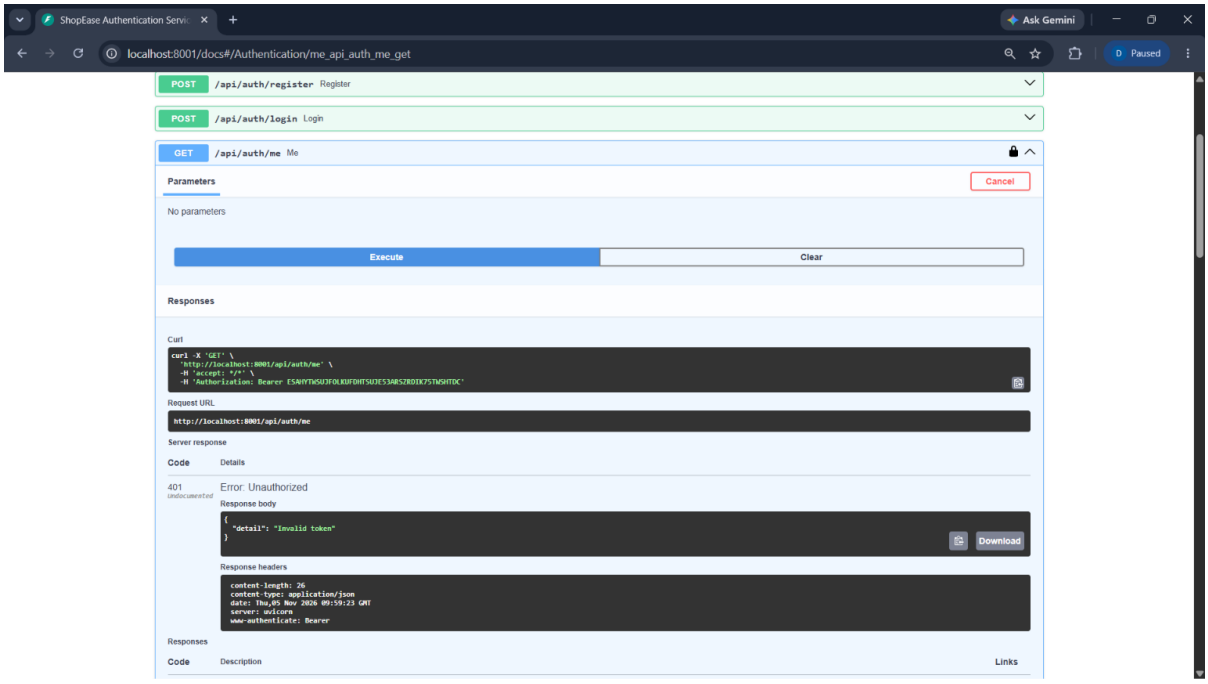
Scenario 8: GET /api/auth/me with Valid Token, 200 OK



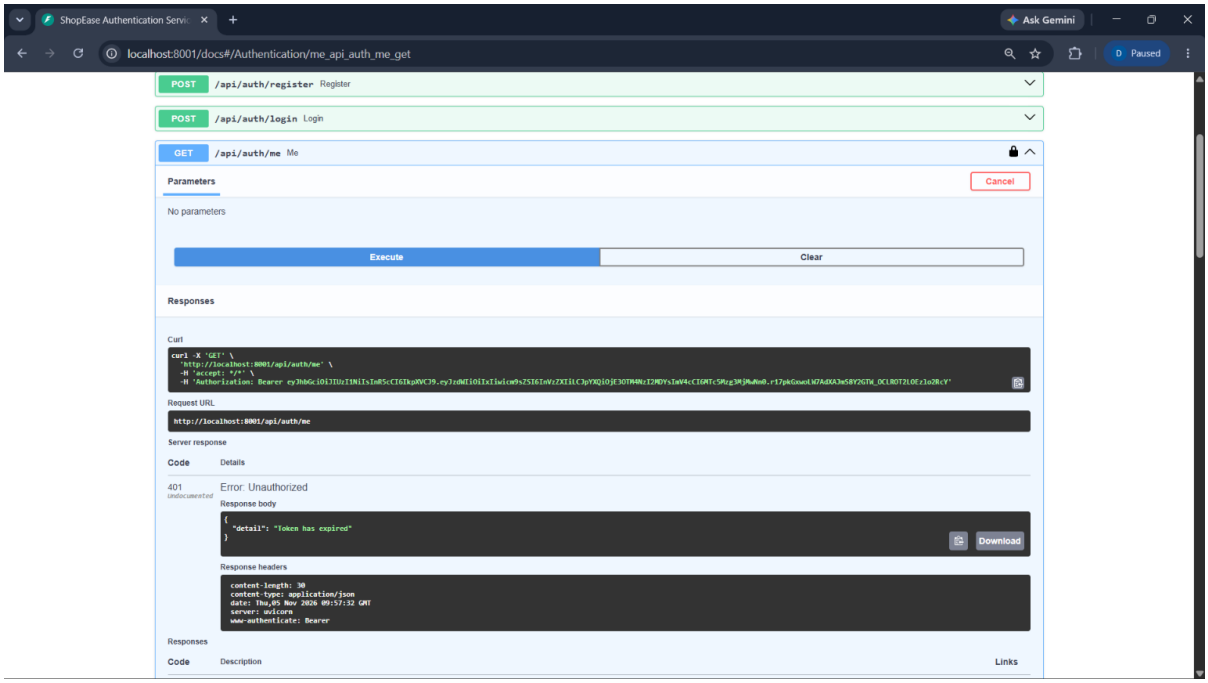
Scenario 9: Protected Endpoint without Token, 401 Unauthorized



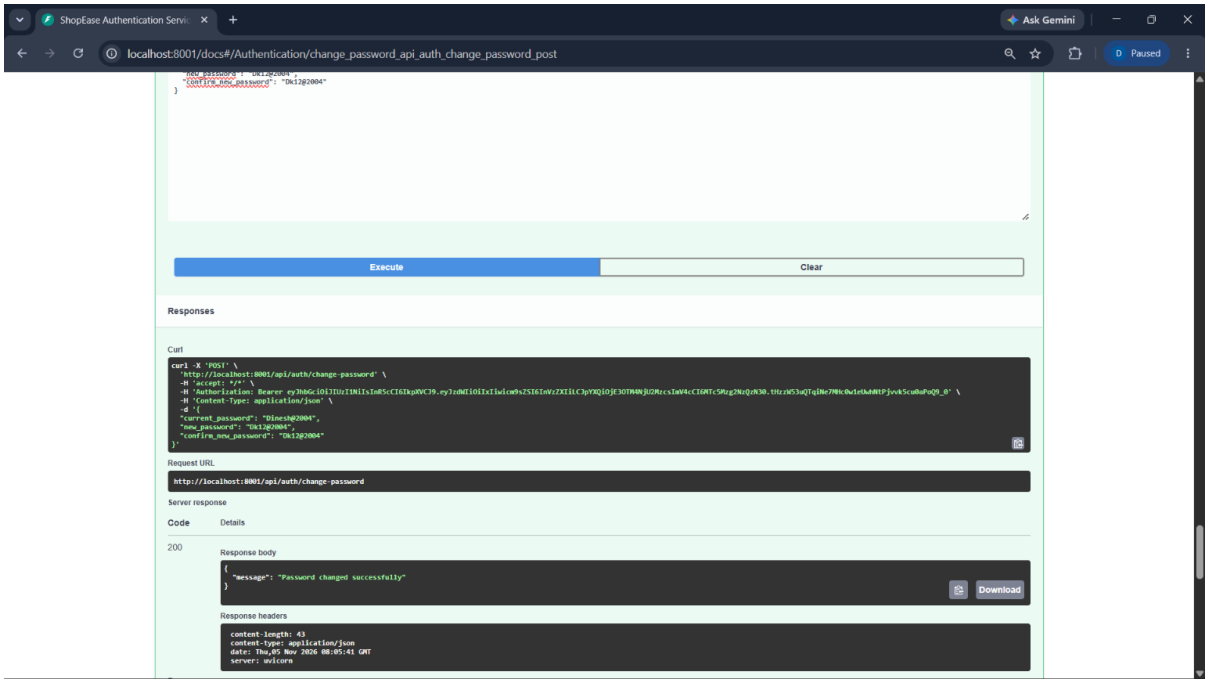
Scenario 10: Protected Endpoint with Invalid Token, 401 Unauthorized



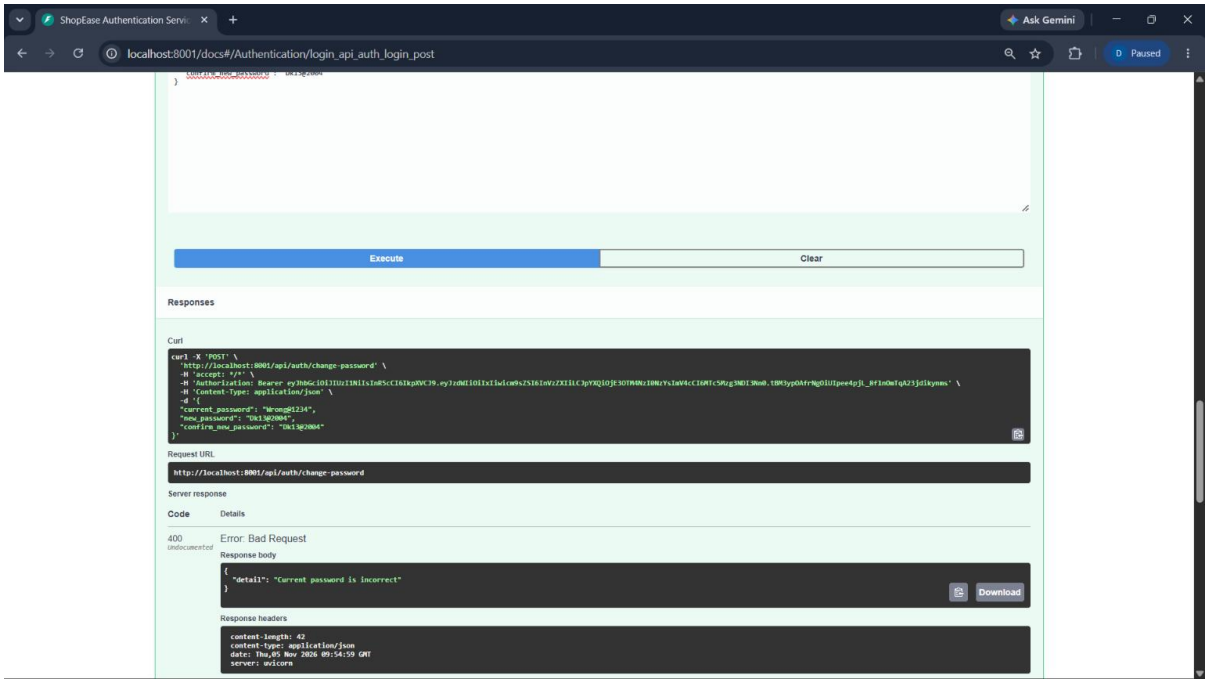
Scenario 11: Protected Endpoint with Expired Token, 401 Unauthorized



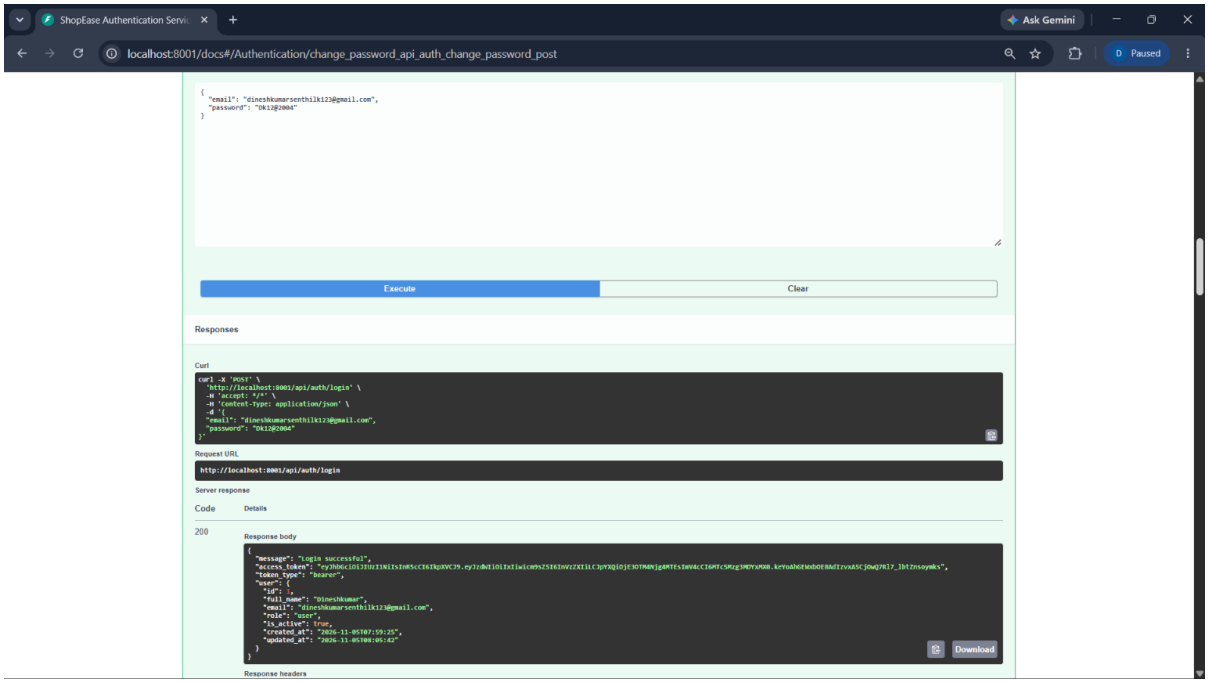
Scenario 12: Change Password with Correct Current Password, 200 OK



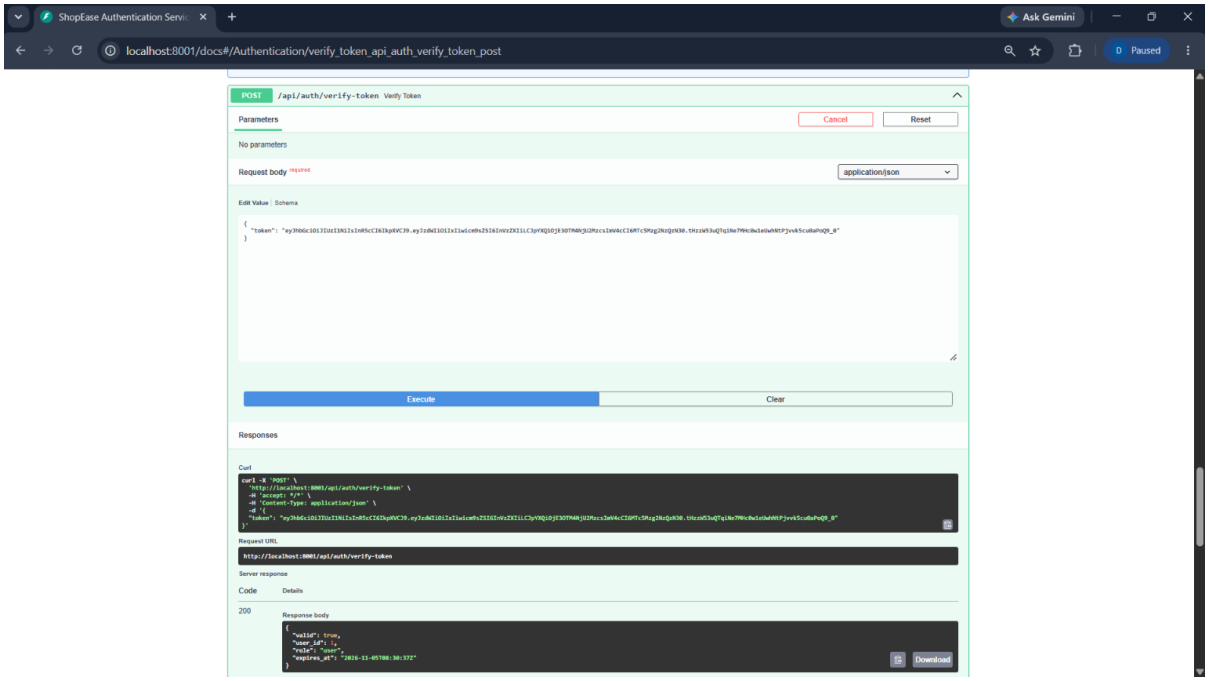
Scenario 13: Change Password with Incorrect Current Password, 400 Bad Request



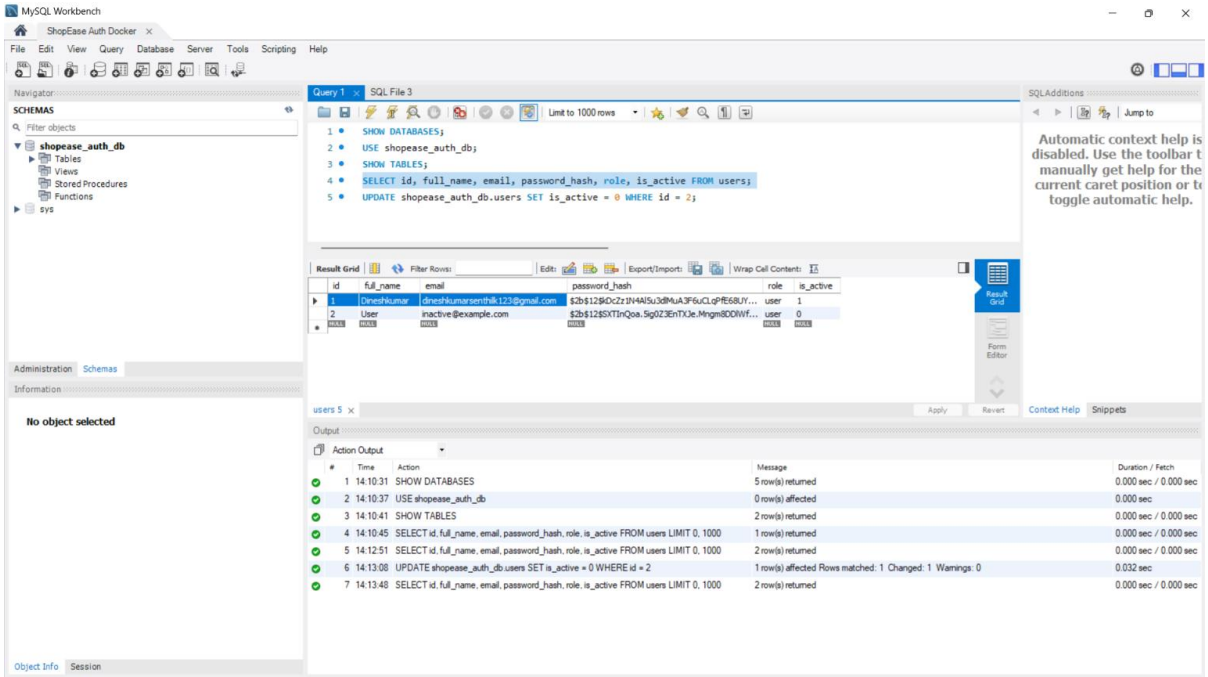
Scenario 14: Login with New Password, 200 OK



Verify Token: POST /api/auth/verify-token, 200 OK



Database: users Table in shopease_auth_db with bcrypt Password Hashes



API Client Testing: Login Request Run from the Imported Collection

